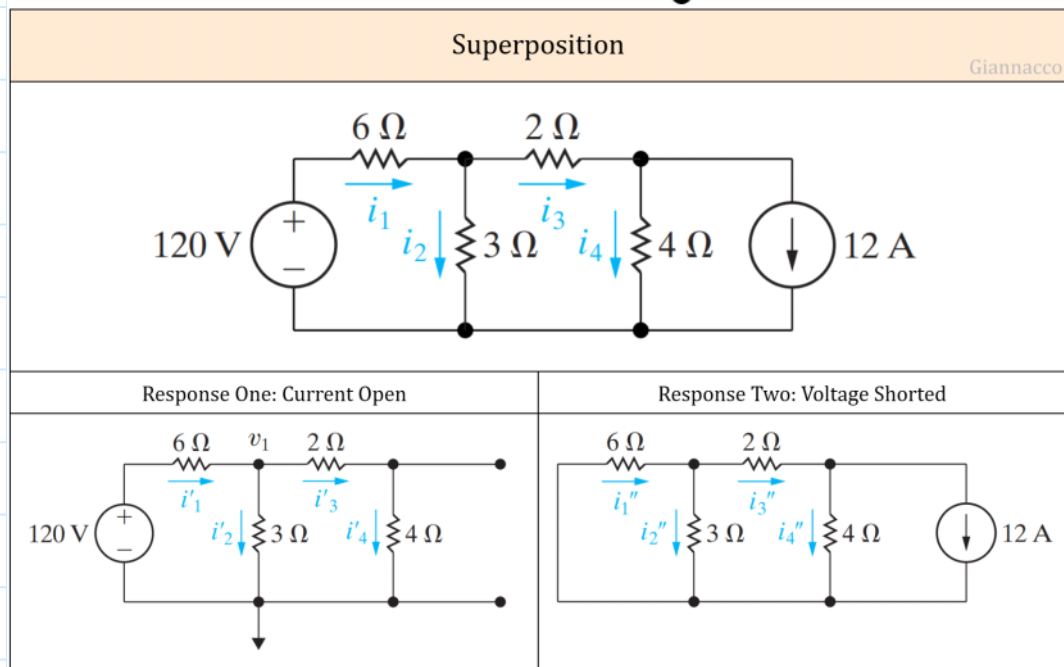
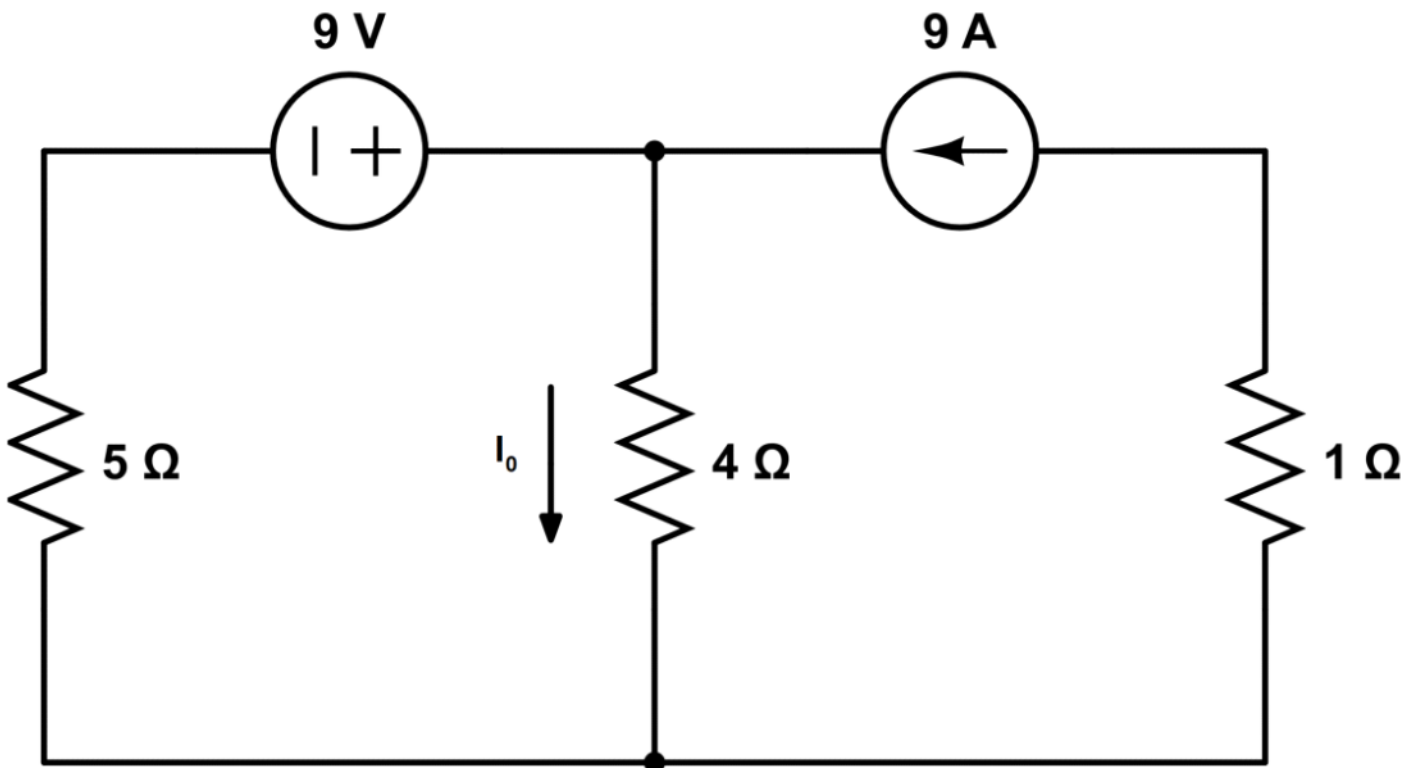


Practice Quiz 3 Solution

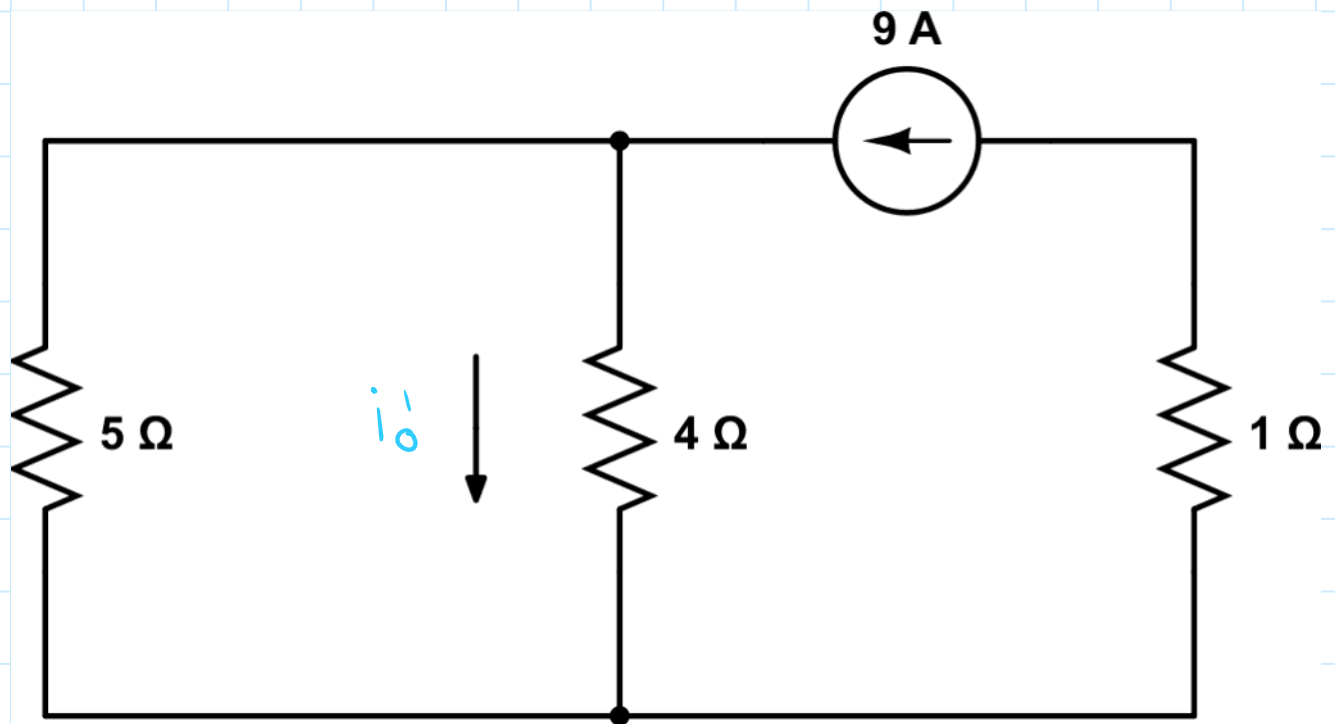
Sunday, September 21, 2025 11:09 PM

For the circuit shown below, use the principle of superposition to find I_0 . Clearly redraw the circuit.



The superposition principle states that the total response of any parameter (voltage or current) in a linear circuit with multiple independent sources is the sum of the individual responses caused by each source acting alone.

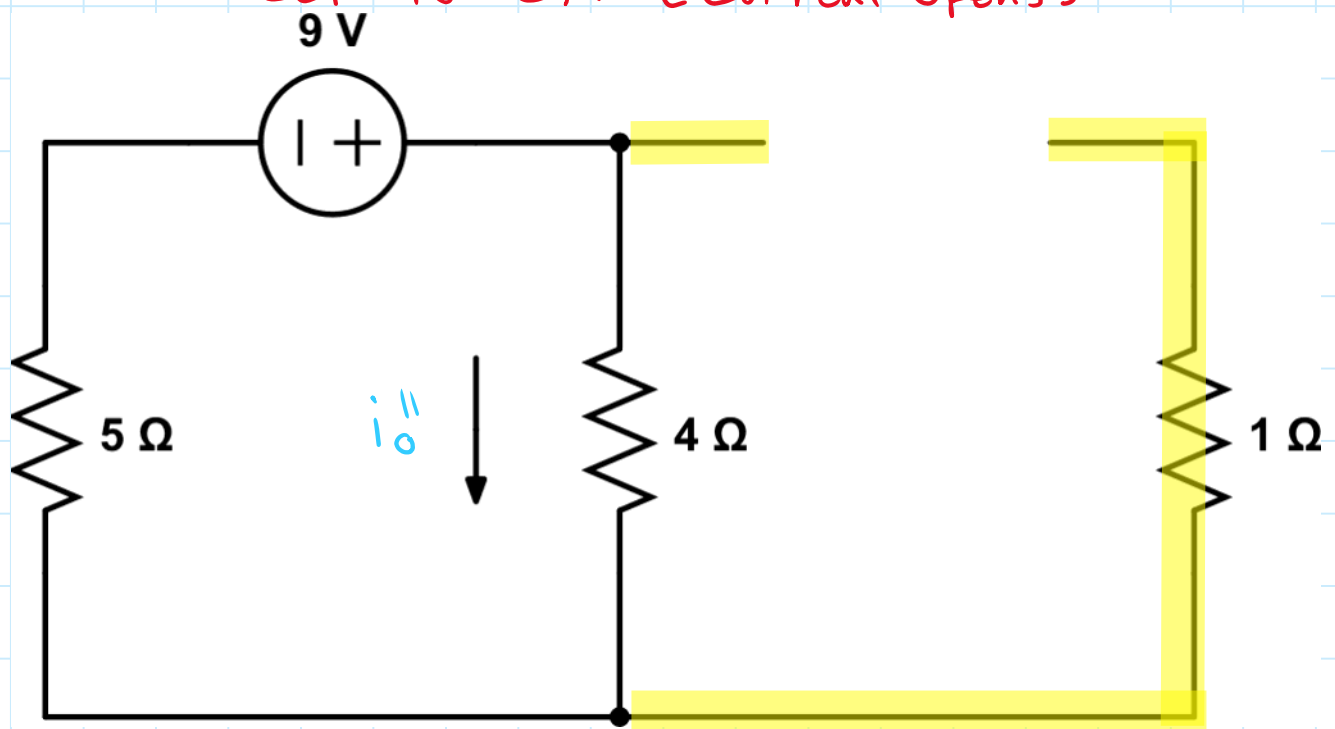
Response One : Independent Voltage Source
set to 0V (Voltage shorts)



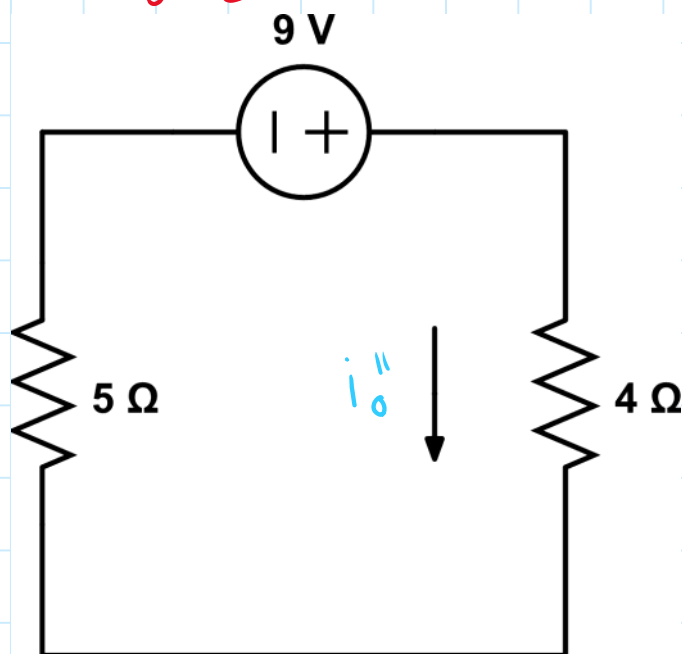
Using Current Division

$$i_o = \frac{\frac{20}{9}\ \Omega}{4\ \Omega} \cdot 9 = 5\text{ A}$$

Response Two: Independent Current Source
set to 0A (Current Opens)



Highlighted areas are an open circuit!



Using KVL:

$$-9 + 4i_o'' + 5i_o' = 0$$

$$9i_o' = 9$$

$$i_o' = 1A$$

$$i_o' + i_o'' = i_o$$

$$5A + 1A = 6A$$

$$i_o = 6A$$